

Google Play Store Exploratory Data Analysis

Comprehensive Technical Report: Dataset Architecture, Preprocessing Pipeline, Statistical Visualizations & Pricing Models

1. Project & Dataset Architecture

This project conducts an end-to-end Exploratory Data Analysis (EDA) on the Google Play Store dataset comprising **10,841 app records** and **13 attributes** [1, 2, 3]. The primary goal is to analyze market dynamics, understand user rating behaviors, quantify monetization models, and identify category-level success factors [8, 16, 18]. The analysis is implemented in Python using pandas, seaborn, matplotlib, and plotly [1, 20].

Column Name	Original Dtype	Non-Null Count	Missing	Description & Processing Note
App	object (str)	10,841	0	App title name string [3]
Category	object (str)	10,841	0	Market category classification (33 distinct) [3, 8]
Rating	float64	9,367	1,474	Average user rating (1.0–5.0); mean-imputed [2, 3, 6]
Reviews	object (str)	10,841	0	User review count; parsed M/K suffixes to Int64 [3, 4]
Size	object (str)	10,841	0	App size in bytes/MB; converted M/K to numeric MB [3, 13]
Installs	object (str)	10,841	1	Download count string; cleaned '+' and ',' to float [3, 4]
Type	object (str)	10,840	1	Monetization type (Free vs Paid) [3, 6, 16]
Price	object (str)	10,841	0	Price in USD string; converted to numeric float [3, 4, 17]
Content Rating	object (str)	10,840	1	Target age rating group [3, 6]
Genres	object (str)	10,841	0	Sub-genre categorization string [3]
Last Updated	object (str)	10,841	1	Last update date string; parsed to datetime [3, 4]
Current Ver	object (str)	10,833	8	Current app version string [3, 5]
Android Ver	object (str)	10,838	3	Minimum supported Android version [3, 5]

2. Data Cleaning & Feature Engineering Pipeline

Raw store data contains significant formatting inconsistencies, string symbols, missing entries, and corrupted rows [4, 5, 6]. A systematic preprocessing pipeline was executed [4, 6]:

- **Symbol Strip & Numeric Coercion:** Cleansed Installs (removed '+' and ',') and converted to numeric format using `pd.to_numeric()` [4].
- **Custom Review Parsing:** Created `parse_reviews()` function to parse 'M' (1,000,000x) and 'K' (1,000x) multipliers into exact integer counts [4].

- **App Size Standardization:** Built `parse_size_to_mb()` converting Megabyte (M) and Kilobyte (K) strings into continuous Megabyte float values [13]. 'Varies with device' was mapped to NaN [13].
- **Missing Value Imputation:** 1,474 missing ratings were imputed with the global rating mean (4.1933) [6]. Missing Installs filled with median, while categorical fields (Type, Content Rating) used mode imputation [6].
- **Outlier & Glitch Correction:** Fixed corrupted Row #10472 (*Life Made Wi-Fi Touchscreen Photo Frame*) where values were shifted due to missing category data, setting Rating from an impossible 19.0 to clean standards [6, 7]. Fixed Row #9148 type value [6, 7].
- **Duplicate Identification:** Uncovered **483 exact duplicate rows** across 10,358 distinct records [7].

```
# Core Data Cleaning Pipeline
df['Installs'] = pd.to_numeric(df['Installs'].str.replace('+', '').str.replace(',', ''), errors='coerce')
df['Rating'] = df['Rating'].fillna(df['Rating'].mean())
df['Size_MB'] = df['Size'].apply(parse_size_to_mb)
paid_apps['Estimated Revenue'] = paid_apps['Price'] * paid_apps['Installs']
```

3. App Category Market Share & Volume Analysis

Analysis of category frequency reveals heavy market concentration in a few dominant software segments [8, 9]. The **FAMILY** category leads with 1,943 apps (17.9%), followed by **GAME** with 1,121 apps (10.3%), and **TOOLS** with 843 apps (7.8%) [9]. Together, the top 3 categories represent over 36% of all store inventory [9].

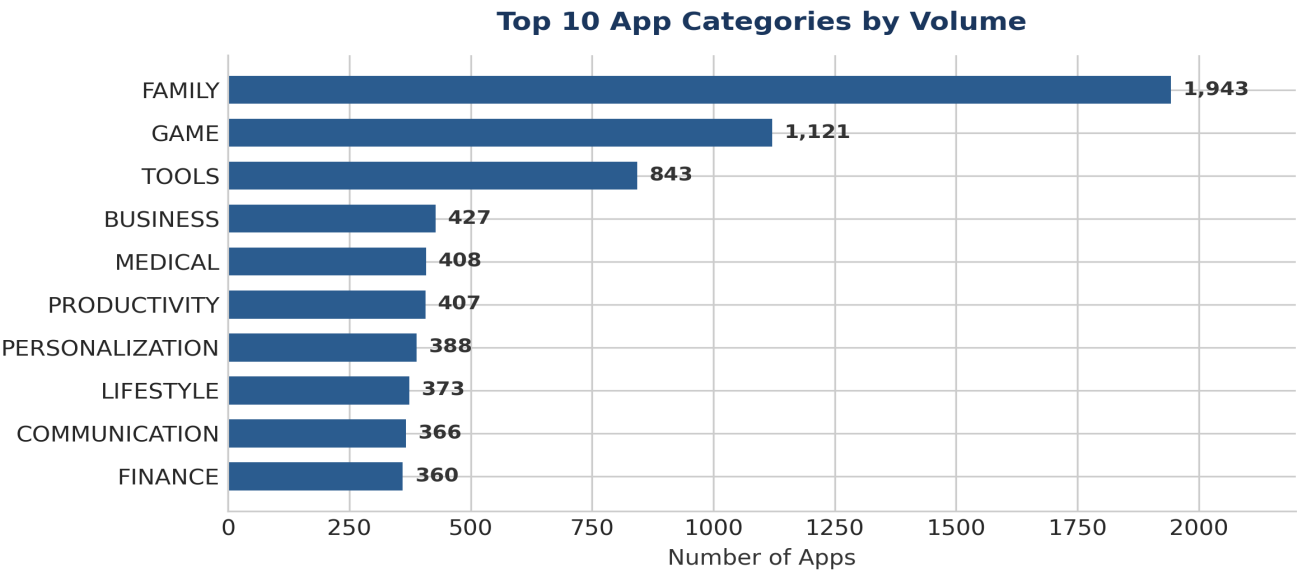


Figure 1: Distribution of top 10 categories by app volume in the Google Play Store dataset [8, 9].

4. User Rating Distribution & Quality Benchmarks

User ratings exhibit a strong negative skewness centered around an overall mean of **4.19** and median of **4.30** [2, 10]. Over 75% of rated apps maintain scores above 4.0 [2, 10]. Category-level mean rating evaluation shows that

EDUCATION (4.37), **EVENTS (4.36)**, and **ART AND DESIGN (4.35)** achieve the highest user satisfaction [11, 12]. Conversely, **DATING (4.01)**, **MAPS AND NAVIGATION (4.07)**, and **TOOLS (4.07)** exhibit the lowest benchmark ratings [11].

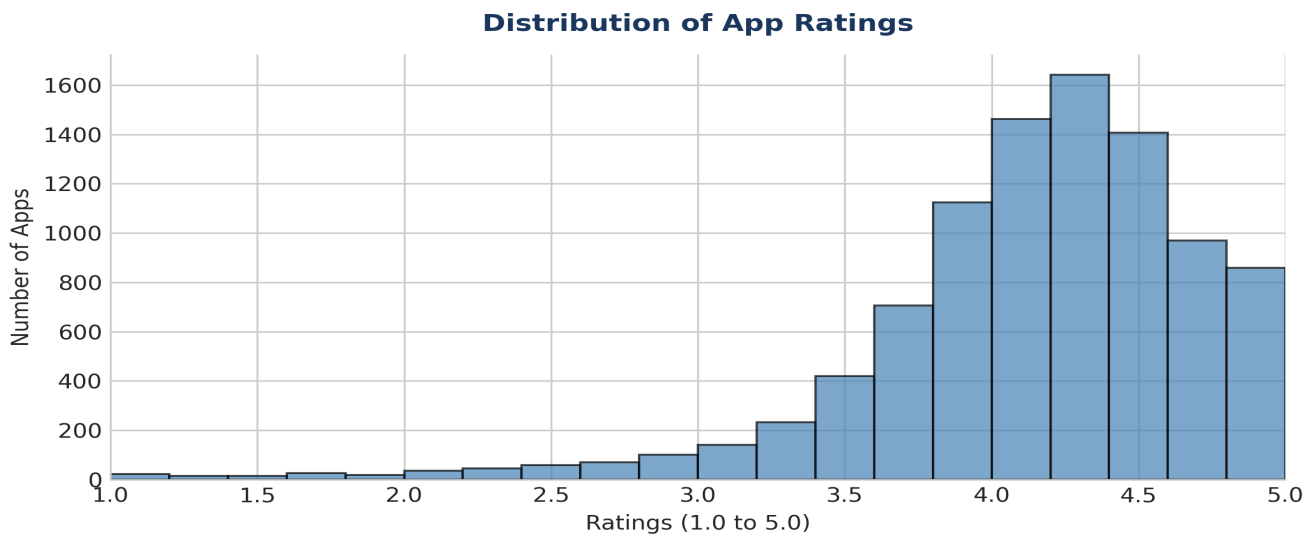


Figure 2: Distribution histogram of app ratings across 10,841 store records [10].

5. App File Size vs. Download Scale Correlation

A critical question for developers is whether larger file sizes hinder download adoption [13]. Calculating Pearson's correlation coefficient between `Size_MB` and `Installs` yields $r = 0.1689$ (~ 0.17) [14, 15]. This indicates a very weak positive relationship [15]. App size alone is *not* a barrier to consumer installation, as high-utility games and productivity tools command massive installs regardless of footprint [13, 15].

TECHNICAL INSIGHT: FILE SIZE IMPACT

Technical Insight (Size Correlation $r=0.17$): App size displays minimal drag on install velocity in modern app marketplaces [15]. Users prioritize functionality, brand reputation, and peer reviews over MB storage constraints [15].

6. Monetization Models & Paid Revenue Analytics

The Play Store is overwhelmingly dominated by the free-to-download model [16]. Free apps constitute **95.91% (9,591 apps)**, while paid apps represent just **4.09% (765 apps)** [16]. Among paid applications, prices range from \$0.99 to \$400.00, with a mean price of \$13.96, a median of \$2.99, and an interquartile range of \$1.49–\$4.99 [17].

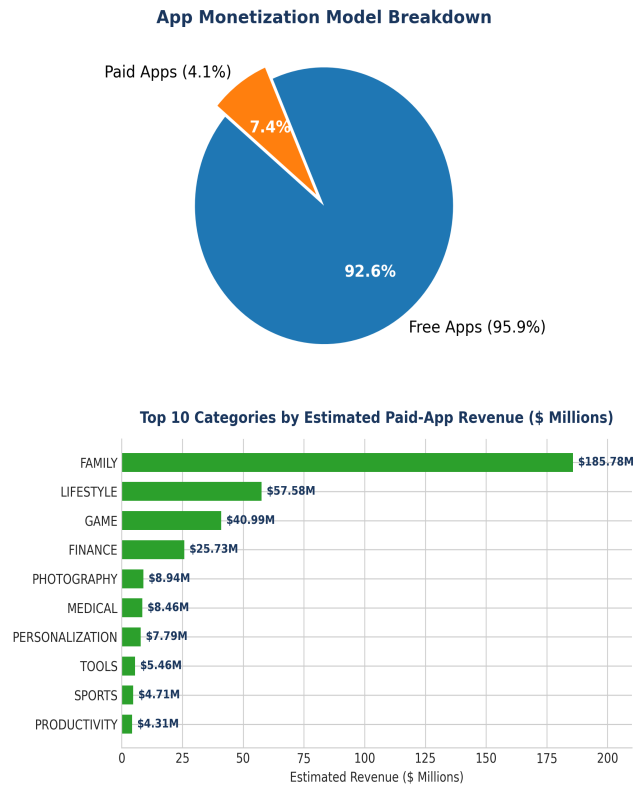


Figure 3: Free vs. Paid app breakdown (left) and Top 10 categories by estimated paid revenue (right) [16, 18].

Estimating gross paid-app revenue (Price × Installs) highlights extreme category concentration [18]:

Category Rank	Category Name	Estimated Gross Revenue (\$)	Revenue Share (%)
1	FAMILY	\$185,780,300	53.6%
2	LIFESTYLE	\$57,583,940	16.6%
3	GAME	\$40,987,640	11.8%
4	FINANCE	\$25,726,680	7.4%
5	PHOTOGRAPHY	\$8,942,768	2.6%
6	MEDICAL	\$8,456,536	2.4%
7	PERSONALIZATION	\$7,786,948	2.2%
8	TOOLS	\$5,464,821	1.6%

7. Popularity (Installs) vs. Rating Quartiles

Direct linear correlation between install volume and user rating is nearly zero ($r = 0.0478$) [18]. However, segmenting applications into four install quartiles using `pd.qcut()` reveals a subtle, crucial non-linear trend [19]. Apps in the **High Install Bucket** average **4.31 rating** compared to **4.07–4.21 rating** in lower install tiers [19, 20]. Top-performing apps benefit from virtuous network effects, where high quality drives installs, and large user bases sustain continuous

developer updates [19].

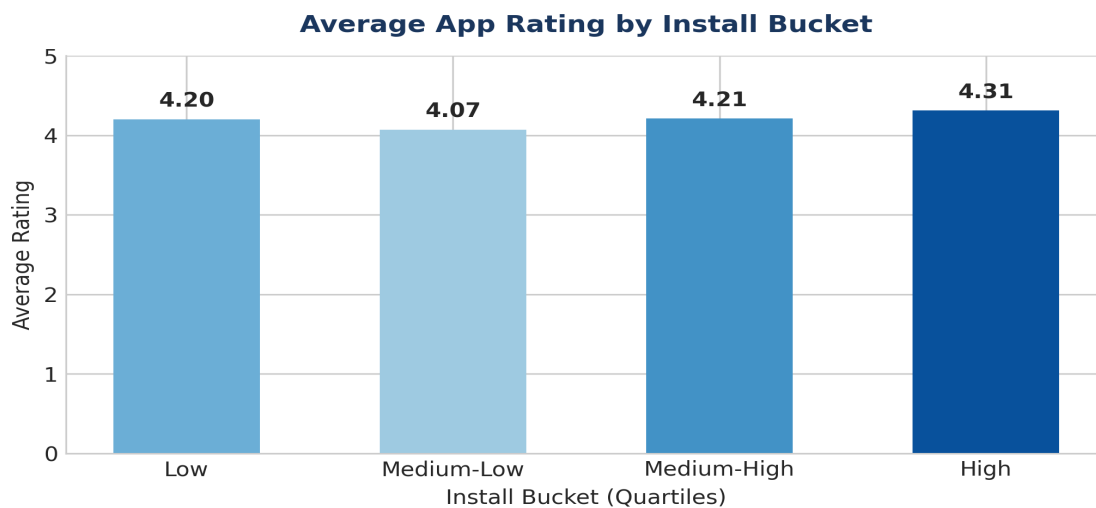


Figure 4: Mean rating across install volume quartiles (Low to High install scale) [19, 20].